



## Doppler sues Bose

Doppler Labs claims Bose backed Kickstarter, took meetings to access technology.  
<http://dgit.in/DLsueBose>



## Pinterest acquires Jelly

The human powered search engine Jelly expands its search and discovery platform.  
<http://dgit.in/PinJelly>

Feature

yourself to the occasional Cyanide & Happiness short, you're unlikely to run into your data cap. However, change in the very nature of content delivery creates problems. With the rise of Netflix and Amazon Prime, millions of people are watching feature-length content in HD, streamed over their internet connections. How many episodes of The Man in the High Castle can you watch until the bandwidth Gestapo throttle your connection to 512 Kbps? People are increasingly ditching TV to watch "TV shows" online, and TV channels themselves are available for streaming. 4G vendors promote their video and music streaming services as value adds to utilize 4G connections. But how can you possibly use Airtel Movies, for instance, if you have a 10 GB monthly data cap? So long as you stick to relatively low resolutions (and if you're frugal with what you stream), a 20-50 GB cap can take you a long way. The arrival of HEVC-encoded content has brought down the bandwidth and data thresholds for streaming good quality video.

But 4K's where it's at, going by media reports. And you're going to run through any data cap in a few days if you try to stream 4K content. One thing we often overlook is that digital TV set-top boxes are piped data, essentially unlimited quantities of it—enough to deliver a 1080p stream of reasonable quality at all times of day, for a fixed price. In that sense, FUP-free data connections do exist. They're just not for services you want to use. FUP limits are fundamentally incompatible with the idea of streaming services, and if the likes of Netflix are to make inroads here in India, more than upping

bandwidth on connections, FUP limits need to either be relaxed or eliminated. The rumour mill suggests that Jio GigaFiber will feature a 10 Mbps FUP-free plan. That is precisely what's needed for a solid streaming experience.

## The Indian Gaming Experience: Drop in, Drop out

We haven't even touched on gaming yet. There are roughly 1 million active Steam users in India, comprising approximately 0.5 percent of the global Steam user base, and a paltry 0.25 percent of the total number of internet users in India. While gaming's huge for us personally, it's most definitely not the average Indian use-case, even by factoring in internet-cafe CS gamers. That being said, the nature of Indian internet connectivity has a significant impact on your gaming experience: Average bandwidth is enough to get the job done with regards to Steam downloads...to an extent. Games frequently exceed 50 GB in size. At 4 Mbps, you're looking at leaving your system on for at least 24 hours. Mandatory updates can leave your games in an effectively unplayable state for quite a bit of time, and that's just the single player experience.

While e-Sports titles like DotA 2 have Indian servers, you'll often have to connect to Asia-Pacific or European servers to play multiplayer in many AAA titles. Here, latency becomes incredibly important. Ping, of course, scales with your distance to the server. Ideal servers are within 1,000 kilometres of your location. When looking at connecting to a European or Australian server, that increases by a factor of ten. When you're looking at a ping in the 300 ms range, instead of competing based on your skill, you're actively fighting latency. With many

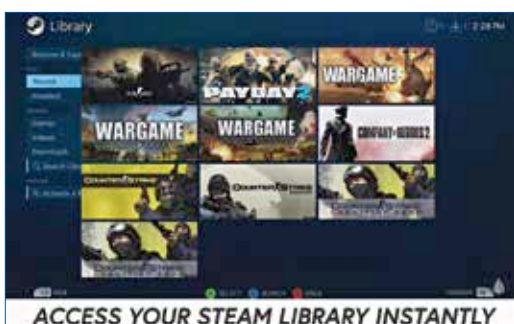


They've got horrible ping and they're losing. Trust us, we just know

AAA games blurring the lines between MMOs and single player experiences, this can turn games like Destiny into nightmares. Latency has absolutely nothing to do with your bandwidth, but it can make or break your multiplayer experience. Cloud game streaming solutions are even more of a reach. For starters, neither GeForce Now nor LiquidSky servers are anywhere near India. When you're streaming the actual gaming experience, latency becomes even more important—anything higher than 100 ms and the illusion that you're playing on local hardware is completely shattered. At present, it's physically impossible to have a decent cloud-streamed gaming experience in India, regardless of bandwidth availability—the servers are just too far away.

## Conclusion

There is, indeed a point where fast becomes too fast. Typical Indian use cases don't actually need more bandwidth than a bog-standard 4 Mbps connection. In more niche areas like gaming and HD streaming, you run into regulatory issues like the FUP, and reliability issues long before bandwidth becomes a constraint. Fiber providers increasingly try to sell us high-bandwidth connections. But when your 100 Mbps fiber connection can only stream a few Netflix episodes before throttling, when you're on the phone with customer service every time there's so much as a strong wind; when you struggle to play multiplayer because of impossible latency, ask yourself: is investing in speed really worth it?



Check it out, something you'll never be able to do!